XELOR AGENT HANDBOOK · BUILT FROM THE IMPLEMENTATION

SPAR Supply

The material availability specialist. Explains what stock exists, what has been ordered from suppliers, and where a shortage threatens a plan or a promise.

Who this is for: product, engineering, operations and anyone who has to explain XELOR to somebody else.
What it covers: the work SPAR reasons about, the rules it cannot break, the exact tools it can call, and what is honestly not built yet.
Truth standard: the repository as it stands on 7 August 2026.

SPAR

01 · AT A GLANCE

What SPAR is for

A role with a fixed tool list, not a general assistant

In one sentence

Explains what stock exists, what has been ordered from suppliers, and where a shortage threatens a plan or a promise.

2	2	4	0
business areas it speaks for	tools it can actually call	capability families allowed	agents it may delegate to

What reaches it

On-hand balances by item and warehouse · Open purchase orders and receipts · Material requirements from AXLE · Component demand from KILN

What it hands back

What is actually available · Where the shortage is, and whose order it threatens · A procurement or stores work item · Traceable references back to the ledger

Capability families it is allowed to touch

inventory. · purchase. · supplier. · agent.

Ownership and authority are not the same thing

The areas above are where SPAR is the right specialist to ask. The tool table on page 8 is the much smaller set it can invoke at runtime. Owning a business area does not grant runtime power over it — and for ONYX, the supervisor, the gap between the two is the widest in the system.

What sits behind SPAR

Records, screens, workflow, controls and hand-offs

Purpose

SPAR module · purchase

Purpose	Controls supplier masters, purchase commitments, approval and receipt of material into the stock ledger.
Core records	Vendors, duplicate candidates, purchase orders/lines/status, W1 approval instances and decisions, goods receipts and receipt references.
User surface	Orders; Vendors; Order detail; Goods receipt. Dashboard evidence comes from the purchase-order register.
End-to-end flow	Create/check vendor; draft purchase order; submit into the configured approval route; permit only the resolved approver at each step; receive no more than the approved open quantity; post the receipt through Inventory inside the same transaction.
Enforced controls	Vendor duplicate evidence, permission and idempotency checks, version-pinned approval, named approver refusal, approved-only receipt, over-receipt prevention and atomic receipt/stock posting.
Inputs and outputs	Receives buy proposals from Planning, consumes Organisation numbering, calls Inventory for receipt, and provides open supply to Planning and working-capital analysis.
Current boundary	Vendor-to-GRN is live. Supplier invoice, three-way match, AP ageing/payment, receipt accruals and supplier performance are not complete.
Primary code map	`apps/web/src/modules/purchase/manifest.ts` · `apps/api/src/modules/purchase/`

How to read these module pages

"Live" means the records, service rules and user/API surface exist in this repository. It does not mean every surrounding enterprise process is complete. The boundary row names what remains illustrative, manual, simulate-driven or outside the MVP.

How SPAR's modules connect

A module owns its records; the agent explains the combined evidence

Inventory

SPAR module · inventory

Purpose	The single source of truth for physical stock movement and current on-hand balance by item, warehouse and batch.
Core records	Warehouses, append-only stock entries, movement type/reference/idempotency key, signed quantity, batch/expiry details and row-locked stock balances.
User surface	Stock; Warehouses. Alerts identify stranded batched stock only when the API can prove it.
End-to-end flow	A domain port asks for a movement; the service validates the request and idempotency key, serialises competing withdrawals, chooses eligible FIFO batches, locks balances, refuses a negative outcome, appends ledger entries and updates balances atomically.
Enforced controls	One write path, advisory and row locks, negative-stock refusal, append-only trigger, movement/reference traceability, FIFO batch consumption and tenant RLS.
Inputs and outputs	Purchase receives, Production issues/receives, Sales dispatches, Quality quarantines/releases and Maintenance consumes/returns spares through this same boundary.
Current boundary	On-hand and the movement ledger are live. No bins, reservations, cycle counting, serial genealogy or full valuation; FIFO chooses the batch, while standard cost remains the valuation basis.
Primary code map	`apps/web/src/modules/inventory/manifest.ts` · `apps/api/src/modules/inventory/`

Cross-module coordination

Purchase creates an approved supply commitment; Inventory records the physical consequence. A receipt is useful only when both succeed in one transaction, and downstream modules move stock through Inventory rather than maintaining their own balances.

Purchase → **Inventory** → **SPAR** bounded evidence → **ONYX / person** decision

How SPAR's world actually runs

The business pipeline, not the agent plumbing

SPAR owns the one thing every other module depends on and none of them may touch: the stock ledger. Five modules move stock, and all five go through one door.

1. **Vendor**
   Created against the same duplicate brain Sales uses — exact code, exact GSTIN, or a trigram name similarity above 0.3 returns 409 with the candidates.

2. **Purchase order**
   Created as a draft, then submitted into W1. The seeded route is two steps: stores review, then admin sign-off. Signing as the wrong one is refused by name.

3. **Goods receipt**
   Only an approved order can be received, and never beyond what remains. The receipt and its stock movement are ONE transaction — stock posts first, so a shortage rolls the whole document back.

4. **The single write path**
   POST /stock/entries, permission inventory.stock.post, Idempotency-Key required. Purchase, Production, Sales, Quality and Maintenance all reach it through a port; none imports Inventory.

5. **The ledger**
   Every movement is appended and signed by quantity. Balances are updated under a row lock, and a negative result is refused rather than recorded.

How much of this SPAR can actually see

Of everything above, SPAR can read exactly one thing directly — stock on hand. The rest of this pipeline is context for judging what it reads; it is not data the agent can reach. Where a mission needs more, it asks the specialist who owns it or it says the evidence is missing.

What the code will not let anyone do

Enforced by constraints and locks, not by wording

Stock cannot go negative

The balance row is locked FOR UPDATE, the delta applied, and a result below zero refused with INSUFFICIENT_STOCK naming what was available.

Withdrawals for one item and warehouse are serialised

An advisory lock is taken before batches are even chosen, so two concurrent issues cannot both plan against the same lot.

The ledger is append-only

A trigger refuses UPDATE and DELETE and the grants are revoked — for the schema owner as well. This is why the demo rebuilds rather than deletes.

Three independent walls, not one

ESLint boundaries fail the build on a cross-module import, the port is a symbol with no concrete class behind it, and the ledger is append-only in the database.

> **Why this page matters more than the agent pages**
> An agent is only as trustworthy as the system it reads. Every rule here holds whether the request came from a person, a screen, a seeder or SPAR — which is precisely why an agent can be given a read tool without also being given a way to do damage.

Where these rules are kept

```
apps/api/src/modules/inventory/inventory.service.ts
```

```
apps/api/src/ports/stock.port.ts
```

```
apps/api/src/modules/purchase/purchase.service.ts
```

How SPAR takes part in a mission

The engine controls order, parallelism and recovery

Person or signal → **ONYX** scope the mission → **SPAR** Supply → **Capability registry** allow-listed tool → **Domain service** business rules → **Tenant data** + audit

1 · Scope

ONYX turns the goal into a run against a frozen graph version, with a fixed tenant, user, step budget and timeout.

2 · Authorise

Two checks, both required: the tool must be on SPAR's allow-list, and the signed-in person must independently hold the permission.

3 · Execute

A registered domain service does the work under the same rules a screen would face. There is no general SQL doorway.

4 · Preserve

Inputs, outputs, events and a checkpoint after every wave, so the run can be explained or resumed.

An agent can narrow authority, never widen it

SPAR is a specialist and delegates to nobody. Because the permission check is repeated against the requesting person, a mission can never reach data that person could not have opened themselves.

06 · EXACT CALLABLE SURFACE

Everything SPAR can call

This table is the complete list — there is no other door

Capability	Mode	What comes back	Permission required	Needs approval
inventory.on-hand.read	Read	Stock on hand	inventory.stock.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

Read · Analyse · Simulate

Return evidence or a reproducible view. Nothing in the business changes.

Draft

Produce reviewable content that is labelled a draft and can never present itself as an approved outcome.

Execute

The only side-effecting mode in the whole system, and the only one that requires an approved human gate above it.

Both locks must open

The agent's allow-list AND the person's permission. Either one failing is a refusal.

What "dispatch" does and does not do

It appends a governed work item naming the responsible specialist, the approval it came from and the exact payload. It does not change a sales order, a stock balance, a production order, a journal or a customer message — a person still does that work.

Where SPAR actually appears

Node by node, from the registered graph definitions

Mission graph	What SPAR does in it
Cross-functional readiness	Reads stock, then assesses supply
Nine-agent operating review	Reads the stock position, then the supply assessment
Controlled action mission	Reads, recommends, and dispatches the shortage-recovery item
Working Capital Review	Reads the stock holding cash
QMS & Audit Readiness	Not involved
Managed Service Assurance	Not involved unless a service issue requires supply evidence

Reading this honestly

"Not involved" is not a limitation, it is the design. A mission asks only the specialists whose evidence it needs, and a specialist cannot add itself to a graph at runtime. The graph is written down, versioned and content-hashed before the run starts.

The shape every mission shares

Goal → Parallel reads → Assessments → Join → HEXA verify → Human gate → Outcome

08 · WORKED EXAMPLE

Buying the metal, two signatures at a time

Real records from the running demo, not an illustration

PO-2627-00003 — 750 kg of SS 316L bright bar from Meridian Metals at ₹2,88,750. The remarks carry the award: Meridian at ₹385/kg against Atlas Alloys at ₹394/kg.

Two different people signed it. The stores in-charge reviewed, the administrator approved, and the trail is hash-chained per instance.

The receipt landed the heat in Pune Stores. After the first tranche was machined, 485 kg of the 750 is still there — and the ledger says exactly where the rest went.

> **The sentence to say out loud**
> “SPAR contributes evidence from its own area, with the source records behind it and its limits stated. It does not claim an action is done until the system has recorded and verified it.”

Fair questions to ask while watching

- Which records is this answer standing on?
- Which permission was checked, and against whom?
- Is this a recorded fact, a simulation, a draft, or a completed result?
- Would this step have waited for a person?
- What happens if the service restarts halfway through?

Live today, and deliberately not

The second column is the one worth reading

Working now

- The single stock write path with FIFO batch consumption
- Append-only ledger with row-locked balances
- Vendor duplicate detection
- Two-step purchase approval through W1
- Gapless per-financial-year document numbering

Not built — stated plainly

- The cycle stops at the receipt — no supplier invoice, no three-way match, no payment, no ageing for direct material
- A goods receipt posts stock but raises no accrual journal
- Valuation is standard cost only; FIFO governs which batch moves, not what it is worth
- No bins below warehouse, no serial tracking, no reservations, no cycle count

Identity boundary	Verified token → tenant and actor
Authority boundary	Agent allow-list AND the person's permission
Action boundary	Side effects need an approved ancestor node
Data boundary	Domain service over a tenant-fenced database
Evidence boundary	Append-only runs, events, checkpoints and audit

Gaps are listed because a guide that hides them fails the first hour of technical due diligence. Every item on the right is either a roadmap decision or a known defect, and both are recorded in the project's own capability-gap register.

SPAR on one page

For explaining the agent to somebody new

Its job

Explains what stock exists, what has been ordered from suppliers, and where a shortage threatens a plan or a promise.

Speaks for

Purchase, Inventory

Takes in

On-hand balances by item and warehouse · Open purchase orders and receipts · Material requirements from AXLE · Component demand from KILN

Gives back

What is actually available · Where the shortage is, and whose order it threatens · A procurement or stores work item · Traceable references back to the ledger

Can call

inventory.on-hand.read, agent.action.dispatch

Cannot do

The cycle stops at the receipt — no supplier invoice, no three-way match, no payment, no ageing for direct material · A goods receipt posts stock but raises no accrual journal · Valuation is standard cost only; FIFO governs which batch moves, not what it is worth

Words used on these pages

Agent	A named specialist with a fixed, allow-listed set of tools — not a process that can roam.
Capability	One registered operation with a declared mode, owner and required permission.
Mission graph	A versioned recipe: which steps run, which run together, where it waits for a person.
Checkpoint	A stored snapshot after each wave, used to explain a run and to resume it safely.
Governed work item	An approved, append-only assignment for a human team — not the business change itself.